Giulia Ischia

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EMPLOYMENT

01/2024 – now — Postdoctoral researcher
Max Planck Institute of Colloids and Interfaces, Germany *Colloid Chemistry Department*
Advisor: Prof. Dr. M. Antonietti.
- Research on biomass conversion into artificial humic substances.
- Supported by an Alexander Von Humboldt fellowship.

05/2022 – 12/2023 — Postdoctoral researcher
University of Trento, Italy *Department of Civil, Environmental and Mechanical Engineering*
Advisor: Prof. L. Fiori.
- Hydrothermal conversion of biomass and bioplastic residues into bio-based products.

04/2018 – 11/2018 — Industrial engineer intern
Dana Corporation, Italia

VISITING RESEARCHER

05/2025 – 06/2025 — Northeast Agricultural University, China (host: prof. F. Yang)
08/2023 – 11/2023 — Tohoku University, Japan (host: prof. T. Adschiri)
03/2021 – 10/2021 — Cornell University, USA (host: prof. J. Goldfarb)
09/2017 – 01/2018 — Boston University, USA (host: prof. J. Goldfarb)

ACADEMIC TEACHING

05/2025 – 06/2025 — Lecturer **Northeast Agricultural University, China**
01/2019 – 07/2023 — Teaching assistant **University of Trento, Italy**

EDUCATION

11/2018 – 05/2022 — PhD in Civil, Environmental and Mechanical Engineering **University of Trento, Italy**
Major: "*Materials, Chemistry and Energy*."
Advisors: Prof. L. Fiori, Prof. A. Miotello.
- Thesis title: "Sustainable conversion of biomass wastes via hydrothermal processes: fundamentals and technology."

09/2015 – 03/2018 — Master of Science in Energy Engineering
University of Trento & Free University of Bozen-Bolzano, Italy
Final grade: 110/110 cum laude.
- Thesis title: "Biofuels production from organic fraction of municipal solid waste through hydrothermal carbonization and pyrolysis" (advisors: prof. L. Fiori, prof. J. Goldfarb).

10/2012 – 07/2015 — Bachelor of Science in Energy Engineering
Politecnico di Torino, Italia

FELLOWSHIP AND AWARDS

7/2024	**Humboldt Research Fellowship**, Alexander von Humboldt Foundation
3/2024	**ACS Sustainability Star**, American Chemical Society (ACS)
7/2023	**International Researcher Visiting Fellowship**, Tohoku University, Japan
6/2021	**Best Young Researcher** WasteEng conference
3/2021	**Fulbright Scholarship**, USA-Italy Fulbright commission
6/2020	**ACS Joseph Breen Memorial Fellowship**, American Chemical Society
8/2020	**The Journey Fellowship**, EIT-KIC Climate

DISSEMINATION AND SCIENTIFIC SERVICES

- **Public engagement**: Science outreach activities within the initiative "Women in Science" for secondary schools.
- **Science communication tools**: Development of project websites for dissemination (ERICSOL, OcchioAlBio!).
- **Peer review:** Reviewer for over 20 international scientific journals including Chemical Engineering Journal, Food chemistry, Waste Management, Nature Communication, Bioresource Technology, Fuel, Journal of Environmental Chemical Engineering.

PRESENTATIONS IN INTERNATIONAL CONFERENCES

2025	**International Symposium on Green Chemistry – ISGC 2025**. *LaRochelle, France*. May. *From biomass to artificial humic substances: a new path for soil health and carbon sequestration.*
2024	**8th Expert Forum on Hydrothermal Processes**. *Leipzig, Germany*. November. *From hydrothermal carbonization to humification for soil carbon sequestration.*
2024	**Green Chemistry (GRS) Gordon Research Seminar**. *Castelldefels, Spain*. June. *From biomass to artificial humic matter for soil carbon sequestration.*
2024	**International Solvothermal and Hydrothermal Synthesis Association seminar series**. Virtual. June. *From biomass to artificial humic matter for soil carbon sequestration.*
2023	**27th Annual Green Chemistry & Engineering Conference**. *Long Beach, USA*. June. *Hydrothermal carbonization as green route for valorizing bioplastic wastes.*
2023	**3rd Symposium on Hydrothermal Carbonization**. *Seoul, South Korea*. May. *Kinetics and chemistry behind the transition from hydrothermal carbonization to liquefaction.*
2022	**American Chemical Society (ACS) Spring**. *Virtual*. February. *From hydrothermal carbonization to liquefaction: The chemistry and kinetics along the hydrothermal biomass reaction spectrum.*
2021	**8th International Conference on Engineering for Waste and Biomass Valorisation**. *Virtual*. June. *Sun to biocoal: a solar - hydrothermal system for waste biomass valorization*. ***Best young researcher award.***
2020	**24th Annual Green Chemistry & Engineering Conference - Systems Inspired Design**. *Virtual*. June. *Post-treatments after hydrothermal carbonization: From organic wastes to solid and liquid biofuels.*
2019	**2nd International Symposium on Hydrothermal Carbonization**, *Berlin, Germany*. May. *Realization of a solar hydrothermal reactor: A hybrid solution to develop a zero-energy technology.*

LIST OF PEER-REVIEWED PUBLICATIONS (* indicates corresponding author)

[19] **Ischia, G.***, Marzban, N., Schmidt, J., Volikov, A. (2026). *Transitioning from hydrothermal carbonization to humification for producing artificial humic substances*. Bioresource Technology. 13336. [10.1016/j.biortech.2025.133306]

[18] **Ischia, G.***, Yoko, A., Wahyudiono, Fiori, L., Adschiri, T. (2026). *Valorization of lignin into carbon spheres, carbon dots, and graphene-like layers via hydrothermal conversion*. International Journal of Biological Macromolecules. 147997. [10.1016/j.ijbiomac.2025.147997]

[17] Zhang, Z., Xie, D., Cheng, K., Liu Z., **Ischia, G.**, Zhao, Y., Yang, F. (2025). *Optimization of Fertilization Strategies for Paddy Fields based on Multi-task Transfer Learning and Reinforcement Learning*. Agricultural Water Management. 322. 109965. [10.1016/j.agwat.2025.109965]

[16] Queffelec, J., Torres Pérez, M.D., Domínguez, H., **Ischia, G.**, Filonenko, S. (2025). *Tailoring the extraction process and properties of polysaccharides from the lichen Evernia prunastri using natural eutectic solvents following a biorefinery approach*. Green Chemistry. 27, 6493–6511. [10.1039/d5gc01413a]

[15] **Ischia, G.**, Marchelli, F., Bazzanella, N., Ceccato, R., Calvi M., Guella, G., Gioia, C., Fiori, L. (2024). *Cellulose Acetates in Hydrothermal Carbonization: a Green Pathway to Valorize Residual Bioplastics*. ChemSusChem, e202401163. [10.1002/cssc.202401163]

[14] **Ischia, G.**, Sudibyo, H., Miotello, A., Tester J.W., Fiori, L., Goldfarb, J.L. (2024). *Identifying the Transition from Hydrothermal Carbonization to Liquefaction of Biomass in a Batch System*. ACS Sustainable Chemistry & Engineering, 12, 4539-4550. [10.1021/acssuschemeng.3c07731]

[13] **Ischia, G.**, Berge, N.D., Bae, S., Marzban, N., Román, S., Farru, G., Wilk, M., Kulli, B., Fiori, L. (2024). *Advances in Research and Technology of Hydrothermal Carbonization: Achievements and Future Directions*. Agronomy 14. [10.3390/agronomy14050955]

[12] **Ischia, G.**, Goldfarb, J.L., Miotello, A., Fiori, L. (2023). *Green solvents to enhance hydrochar quality and clarify effects of secondary char*. Bioresource Technology, 129724. [10.1016/j.biortech.2023.129724]

[11] **Ischia, G.**, Marchelli, F., Fiori, L. (2023). Advancing bioenergy for a green future. Detritus, 23, 1–2. [10.31025/2611-4135/2023.18278]

[10] Marchelli, F., Ferrentino, R., **Ischia, G.**, Calvi, M., Andreottola, G., Fiori, L. (2023). *Valorisation of eyewear bioplasics through HTC and anaerobic digestion: preliminary results*. Detritus, 23, 35–42. [10.31025/2611-4135/2023.18275]

[9] Marchelli, F., **Ischia, G.**, Calvi, M., Fiori, L. (2023). *Hydrothermal Carbonization as a Process to Facilitate the Disposal of Bioplastics*. Chemical Engineering Transactions, 99, 79–84. [10.3303/CET2399014]

[8] Battocchio, P., Bazzanella, N., Orlandi, M., **Ischia, G.**, Miotello, A. (2023). *Carbon nanoparticles as absorbers in PVC for laser ablation propulsion: size effects*. Applied Physics A: Materials Science and Processing, 129, 1–12. [10.1007/s00339-023-06502-7]

[7] **Ischia, G.**, Cutillo, M., Guella, G., Bazzanella, N., Cazzanelli, M., Orlandi, M., Miotello, A., Fiori, L. (2022). *Hydrothermal carbonization of glucose: Secondary char properties, reaction pathways, and kinetics*. Chemical Engineering Journal, 449, 137827.[10.1016/j.cej.2022.137827]

[6] **Ischia, G.***, Cazzanelli, M., Fiori, L., Orlandi, M., Miotello, A. (2022): *Exothermicity of hydrothermal carbonization: Determination of heat profile and enthalpy of reaction via high-pressure differential scanning calorimetry*. Fuel, 310, 122312. [10.1016/j.fuel.2021.122312]

[5] Gonnella, G., **Ischia, G.**, Fambri, L., Fiori, L. (2022). *Thermal Analysis and Kinetic Modeling of Pyrolysis and Oxidation of Hydrochars*. Energies, 15, 1–21. [10.3390/en15030950]

[4] **Ischia, G.**, Fiori, L. (2021): *Hydrothermal Carbonization of Organic Waste and Biomass. A Review on Process, Reactor, and Plant Modeling.* Waste and Biomass Valorization, 12, 2797–2824. [10.1007/s12649-020-01255-3]. **Most cited article of the journal in 2022.**

[3] **Ischia, G.**, Fiori, L., Gao, L., Goldfarb, J.L. (2021). *Valorizing municipal solid waste via integrating hydrothermal carbonization and downstream extraction for biofuel production*. Journal of Cleaner Production, 289, 125781. [10.1016/j.jclepro.2021.125781]

[2] **Ischia, G.**, Castello, D., Orlandi, M., Miotello, A., Rosendahl, L.A., Fiori, L. (2020). *Waste to biofuels through zero-energy hydrothermal solar plants: Process design.* Chemical Engineering Transactions, 80, 7–12. [10.3303/CET2080002]

[1] **Ischia, G.**, Orlandi, M., Fendrich, M.A., Bettonte, M., Merzari, F., Miotello, A., Fiori, L. (2020). *Realization of a solar hydrothermal carbonization reactor: A zero-energy technology for waste biomass valorization*. Journal of Environmental Management, 259. [10.1016/j.jenvman.2020.110067]